Clock Synchronization and Vector Clocks in Distributed Systems

In distributed systems without shared memory or a single synchronized physical clock, ordering events across independent processes represents a fundamental challenge. Physical quartz clocks suffer from clock drift caused by temperature fluctuations and hardware imperfections. Network Time Protocol (NTP) synchronizes clocks across packet networks via statistical jitter filtering, but bounds cannot guarantee strict global ordering for causal consistency.

Leslie Lamport introduced logical clocks where each process maintains a monotonically increasing integer counter. If event A causally precedes event B (happened-before relation, denoted A -> B), then L(A) < L(B). However, the converse does not hold: L(A) < L(B) does not imply causal relationship. Vector Clocks resolve this deficiency by having each node maintain a vector of clock values for all nodes in the cluster. Incrementing the local index on internal events and transmitting the vector with every message allows nodes to detect concurrent operations, establish partial orders, and resolve split-brain conflicts.